

Latest update: 09/28, 18:00

Since I forgot that this course has international students, all assignment requirements, from now on, will be written entirely in English.

The requirements below may slightly conflict with project2.pdf. If you find any conflicts, please email me at M11515013@mail.ntust.edu.tw, and I may make corrections.

Assignment requirements: For security reasons, all submitted assignments will be executed inside a VM on the server, where they will be automatically graded, scored, and performance-tested. The environment is as follows:

1. Ubuntu 24.04
2. Compiler: GCC, standard C++17 recommended
3. 8GB RAM
4. Execution time limit per test case: 60 seconds
5. Build time limit: 60 seconds
6. No network access inside the VM

But don't worry, if you write your code normally and only use the C++ standard library, the OS, compiler, and RAM should not be an issue. Execution and compilation should finish within seconds. They should not take several minutes.

Required directory structure: After building and running, the project should ideally look like this:

M123123123/

```
.git/          # optional, if you use git
results/       # directory for output files outputXXX.cbi
build/         # cmake build directory
inputs/        # directory for test data
src/           # source code
CMakeLists.txt
README.md
```

At submission, the project should ideally look like this:

M123123123/

```
.git/          # optional, if you use git
src/           # source code
CMakeLists.txt
```

README.md

Submission notes:

1. Do not upload the **build/**, **results/**, or **inputs/** directories. The server will build and run the program on its own.
2. The project directory must be named after your student ID and compressed as **StudentID.zip**. ex: M11515013.zip
3. Input files are named input001.cbi, input002.cbi, ... (three digits). Output file names must match, i.e., output001.cbi, output002.cbi. (A file name like input1.cbi is invalid; it must use three digits. The professor's project2.pdf uses input1.cbi, but please follow the rule stated here.)
4. In addition to the program, you must submit a report in PDF format. .md and .docs files are not accepted. Name the report after your student ID and upload it as **StudentID.pdf** together with **StudentID.zip**. **Do not put the report inside StudentID.zip.**
5. The compiled program must be named **cbi**. You can specify the name in CMakeLists.txt.
6. The first two arguments of the cbi program must be the input file path and the output file path, respectively. If not enough arguments are provided, the program must report an error message. Relative paths must be supported, for example:

correct

```
cbi input001.cbi output001.cbi
```

correct

```
cbi ../abc123/input001.cbi ../def456/output001.cbi
```